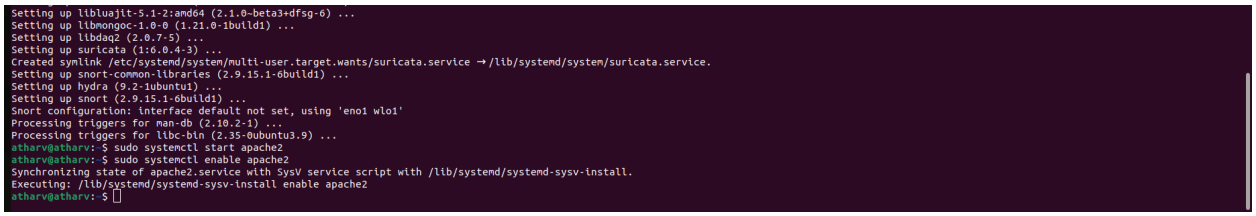
Procedure/Program:

Step 1: Update Ubuntu and Install Required Packages

1. Update the package repository:
sudo apt update && sudo apt upgrade -y

2. Install security tools:
sudo apt install snort suricata nmap nikto hydra apache2 -y

3. Enable Apache web server:
sudo systemctl start apache2
sudo systemctl enable apache2



Part 1: Configuring Intrusion Detection Systems (IDS)

Step 2: Configure Snort IDS

1. Verify Snort Installation:
snort -V

2. Edit Snort Configuration File:
sudo nano /etc/snort/snort.conf
Set the HOME_NET to match your network:
ipvar HOME_NET 10.10.10.0/24

preprocessor sfportscan: proto { all } scan_type { all } sense_level { low }
output alert_fast: /var/log/snort/alert

3. Run Snort in IDS Mode:
sudo snort -A console -c /etc/snort/snort.conf -i eth0
(eth0: interface name, check ifconfig for the interface name and IP address)

```
4057 Snort rules read
    3383 detection rules
    0 decoder rules
    0 preprocessor rules
3383 Option Chains linked into 949 Chain Headers
+++++++++++++++++++++++++++++++++++++++++++++++++++++++++++

+-------------------[Rule Port Counts]--------------------------------------
|             tcp     udp    icmp      ip
|     src     151      18       0       0
|     dst    3306     126       0       0
|     any     383      48      52      22
|      nc      27       8      15      20
|     s+d      12       5       0       0
+----------------------------------------------------------------------------

+-----------------------[detection-filter-config]-----------------------------
| memory-cap : 1048576 bytes
+-----------------------[detection-filter-rules]------------------------------
| none
-----------------------------------------------------------------------------

+-----------------------[rate-filter-config]----------------------------------
| memory-cap : 1048576 bytes
+-----------------------[rate-filter-rules]-----------------------------------
| none
-----------------------------------------------------------------------------

+-----------------------[event-filter-config]---------------------------------
| memory-cap : 1048576 bytes
+-----------------------[event-filter-global]---------------------------------
| none
+-----------------------[event-filter-local]----------------------------------
| gen-id=1      sig-id=1991       type=Limit     tracking=src count=1   seconds=60
| gen-id=1      sig-id=2923       type=Threshold tracking=dst count=10  seconds=60
| gen-id=1      sig-id=2495       type=Both      tracking=dst count=20  seconds=60
| gen-id=1      sig-id=2924       type=Threshold tracking=dst count=10  seconds=60
| gen-id=1      sig-id=3152       type=Threshold tracking=src count=5   seconds=2
| gen-id=1      sig-id=2496       type=Both      tracking=dst count=20  seconds=60
| gen-id=1      sig-id=2275       type=Threshold tracking=dst count=5   seconds=60
| gen-id=1      sig-id=2494       type=Both      tracking=dst count=20  seconds=60
| gen-id=1      sig-id=2523       type=Both      tracking=dst count=10  seconds=10
| gen-id=1      sig-id=3273       type=Threshold tracking=src count=5   seconds=2
+-----------------------[suppression]----------------------------------------
| none
-----------------------------------------------------------------------------
Rule application order: pass->drop->sdrop->reject->alert->log
Verifying Preprocessor Configurations!
WARNING: flowbits key 'smb.tree.create.llsrpc' is set but not ever checked.
WARNING: flowbits key 'ms_sql_seen_dns' is checked but not ever set.
33 out of 1024 flowbits in use.

[ Port Based Pattern Matching Memory ]
+- [ Aho-Corasick Summary ] -----------------------------------
| Storage Format    : Full-Q
```

```
| memory-cap : 1048576 bytes
+-----------------------[event-filter-global]---------------------------------
| none
+-----------------------[event-filter-local]----------------------------------
| gen-id=1      sig-id=1991       type=Limit     tracking=src count=1   seconds=60
| gen-id=1      sig-id=2923       type=Threshold tracking=dst count=10  seconds=60
| gen-id=1      sig-id=2495       type=Both      tracking=dst count=20  seconds=60
| gen-id=1      sig-id=2924       type=Threshold tracking=dst count=10  seconds=60
| gen-id=1      sig-id=3152       type=Threshold tracking=src count=5   seconds=2
| gen-id=1      sig-id=2496       type=Both      tracking=dst count=20  seconds=60
| gen-id=1      sig-id=2275       type=Threshold tracking=dst count=5   seconds=60
| gen-id=1      sig-id=2494       type=Both      tracking=dst count=20  seconds=60
| gen-id=1      sig-id=2523       type=Both      tracking=dst count=10  seconds=10
| gen-id=1      sig-id=3273       type=Threshold tracking=src count=5   seconds=2
+-----------------------[suppression]----------------------------------------
| none
-----------------------------------------------------------------------------
Rule application order: pass->drop->sdrop->reject->alert->log
Verifying Preprocessor Configurations!
WARNING: flowbits key 'smb.tree.create.llsrpc' is set but not ever checked.
WARNING: flowbits key 'ms_sql_seen_dns' is checked but not ever set.
33 out of 1024 flowbits in use.

[ Port Based Pattern Matching Memory ]
+- [ Aho-Corasick Summary ] -----------------------------------
| Storage Format    : Full-Q
| Finite Automaton  : DFA
| Alphabet Size     : 256 Chars
| Sizeof State      : Variable (1,2,4 bytes)
| Instances         : 215
|     1 byte states : 204
|     2 byte states : 11
|     4 byte states : 0
| Characters        : 64755
| States            : 31951
| Transitions       : 863868
| State Density     : 10.6%
| Patterns          : 5041
| Match States      : 3836
| Memory (MB)       : 16.90
|   Patterns        : 0.51
|   Match Lists     : 1.01
|   DFA
|     1 byte states : 1.02
|     2 byte states : 13.96
|     4 byte states : 0.00
+-[ Number of patterns truncated to 20 bytes: 1038 ]----------------------
pcap DAQ configured to passive.
Acquiring network traffic from "eth0".
Reload thread starting...
Reload thread started, thread 0x70a8f0800640 (54393)
ERROR: Can't start DAQ (-1) - SIOCGIFHWADDR: No such device!
Fatal Error, Quitting..
atharv@atharv: $
```

Part 2: Performing Vulnerability Analysis

Step 3: Scan for Open Ports Using Nmap

1. Identify the Target IP Address:

   ip a

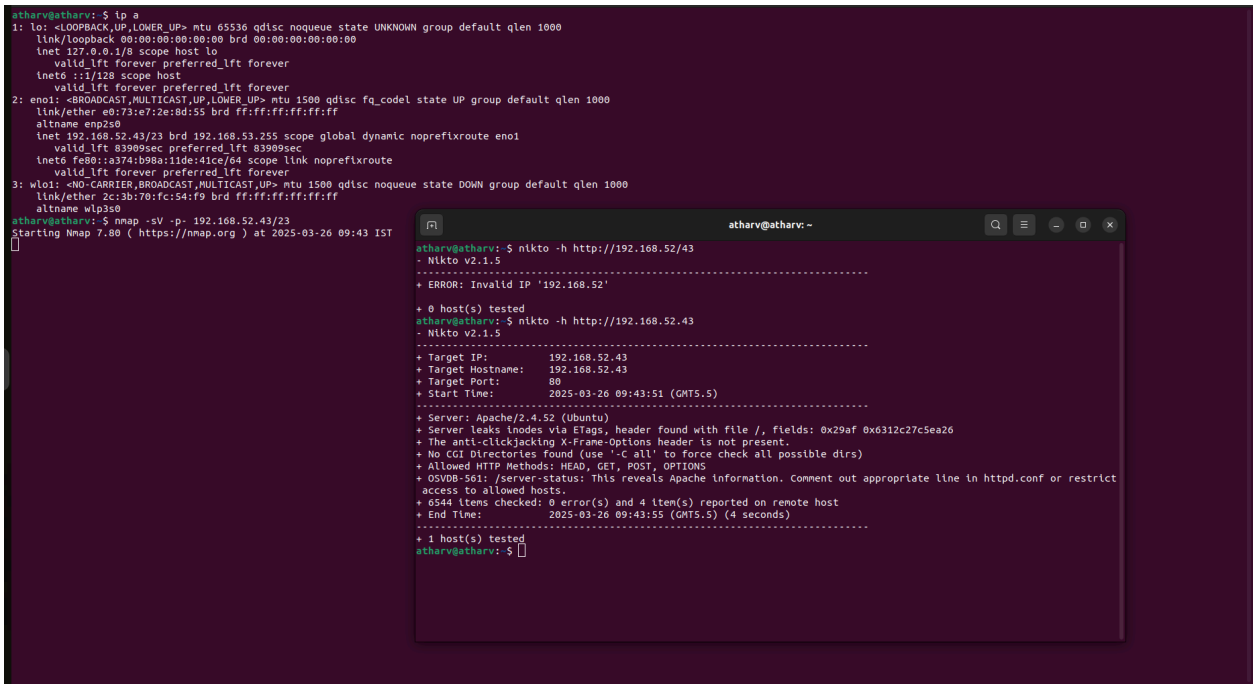
2. Run Nmap Scan:

   nmap -sV -p- 10.10.10.6 (Check for your system IP address here)

Step 4: Web Vulnerability Scan Using Nikto

1. Run Nikto Scan:

nikto -h http://10.10.10.6 (Check for your system IP address here)



Part 3: Performing Vulnerability Analysis

Step 5: Simulate Port Scanning Attack

1. Run an aggressive Nmap scan:
   nmap -A 10.10.10.6 (Check for your system IP address here)

2. Check Snort Logs:
   sudo cat /var/log/snort/alert

```
atharv@atharv:~$ ps aux | grep snort
snort      53020  0.0  0.9 200652 153920 ?       Ssl  09:26   0:00 /usr/sbin/snort -m 027 -D -d -l /var/log/snort -u snort -g snort --pid-path /run/snort/ -c /etc/snort/snort.conf -S "HOME_NET=[192.168.0.
0/16]" -i eno1
snort      53035  0.0  0.5 131616 84632 ?        Ssl  09:26   0:00 /usr/sbin/snort -m 027 -D -d -l /var/log/snort -u snort -g snort --pid-path /run/snort/ -c /etc/snort/snort.conf -S "HOME_NET=[192.168.0.
0/16]" -i wlo1
atharv     55821  0.0  0.0   9532  2560 pts/0    S+   09:49   0:00 grep --color=auto snort
atharv@atharv:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eno1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether e0:73:e7:2e:8d:55 brd ff:ff:ff:ff:ff:ff
    altname enp2s0
    inet 192.168.52.43/23 brd 192.168.53.255 scope global dynamic noprefixroute eno1
       valid_lft 83437sec preferred_lft 83437sec
    inet6 fe80::a374:b98a:11de:41ce/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: wlo1: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default qlen 1000
    link/ether 2c:3b:70:fc:54:f9 brd ff:ff:ff:ff:ff:ff
    altname wlp3s0
atharv@atharv:~$ nmap -Pn -A 192.168.52.43
Starting Nmap 7.80 ( https://nmap.org ) at 2025-03-26 09:50 IST
Stats: 0:00:56 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 91.67% done; ETC: 09:51 (0:00:01 remaining)
Stats: 0:01:12 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 91.67% done; ETC: 09:51 (0:00:02 remaining)

atharv@atharv:~$ nmap -A 10.10.10.6
Starting Nmap 7.80 ( https://nmap.org ) at 2025-03-26 09:52 IST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.21 seconds
atharv@atharv:~$ nmap -A 192.168.53.43
Starting Nmap 7.80 ( https://nmap.org ) at 2025-03-26 09:52 IST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.21 seconds
atharv@atharv:~$ nmap -A 192.168.52.43
Starting Nmap 7.80 ( https://nmap.org ) at 2025-03-26 09:52 IST
Stats: 0:01:02 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 91.67% done; ETC: 09:53 (0:00:02 remaining)

atharv@atharv:~$ sudo cat /var/log/snort/alert
cat: /var/log/snort/alert: No such file or directory
atharv@atharv:~$
```

Step 6: Simulate Port Scanning Attack

1. Run Hydra Brute-Force Attack on SSH:
hydra -L users.txt -P passwords.txt ssh://10.10.10.6

(Please create the user.txt and passwords.txt file by using mkdir command then Add some sample user as “test” and password as “toor” on the 10.10.10.6 system. Perform this attack from another VM like Kali Linux with ip as 10.10.10.4. Please remember all the IPs given may vary from system to system)

2. Check Snort Rules:
sudo nano /etc/snort/snort.conf
include $RULE_PATH/local.rules (Look for this line in the configuration file)
sudo nano /etc/snort/rules/local.rules (Edit this file)

alert tcp any any -> $HOME_NET any (msg:"Port scan detected"; flags:S; threshold: type threshold, track by_src, count 10, seconds 60; sid:1000001; rev:1;)

alert tcp any any -> $HOME_NET 22 (msg:"SSH brute force attack detected"; flow:to_server,established; detection_filter:track by_src, count 5, seconds 30;

sid:1000002; rev:1;)
(Add these 2 rules in the local.rules file)

3. Check Snort Logs:
sudo cat /var/log/snort/alert

```
[DATA] max 4 tasks per 1 server, overall 4 tasks, 9 login tries (l:3/p:3), ~3 tries per task
[DATA] attacking ssh://192.168.0.0:22/
[ERROR] could not connect to ssh://192.168.0.0:22 - Timeout connecting to 192.168.0.0
atharv@atharv:~/Documents/IS$ sudo nano /etc/snort/snort.conf
[sudo] password for atharv:
atharv@atharv:~/Documents/IS$ sudo nano /etc/snort/rules/local.rules
atharv@atharv:~/Documents/IS$ sudo nano /etc/snort/rules/local.rules
atharv@atharv:~/Documents/IS$ sudo cat /var/log/snort/alert
03/26-10:07:54.110227  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff15:e740
03/26-10:09:09.070067  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff9e:ac6b
03/26-10:09:17.721530  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff88:aaa
03/26-10:09:55.601856  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ffba:18be
03/26-10:09:56.137214  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:09:59.817696  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:09:59.823467  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:10:15.699575  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff5c:c4db
03/26-10:10:16.471064  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:10:18.674477  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff49:62bb
03/26-10:10:20.470464  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:10:20.474924  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:10:40.651393  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:10:41.122069  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:10:41.563762  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff9e:ac6b
03/26-10:10:41.769862  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:10:42.063731  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff9e:ac6b
03/26-10:11:01.428111  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff5a:275
03/26-10:11:06.074387  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff61:f1bf
03/26-10:11:15.319897  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:11:15.323919  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:11:17.174464  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ffeb:4253
03/26-10:11:23.072880  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:11:23.076822  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:11:28.348770  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:11:35.753860  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:11:36.122766  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff1c:746f
03/26-10:11:38.226077  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:11:39.135898  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::16
03/26-10:11:39.464188  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::16
03/26-10:11:39.572238  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:11:39.620188  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ffbc:2d8c
03/26-10:11:39.664016  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:11:49.265046  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:11:49.449937  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:11:49.670468  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff49:62bb
03/26-10:11:50.226882  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:11:50.678604  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff49:62bb
03/26-10:12:31.598159  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ffd7:f8e7
03/26-10:12:31.599387  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::16
03/26-10:12:31.816478  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::16
03/26-10:12:32.725350  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff5c:c4db
03/26-10:12:33.723483  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:13:26.297169  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:13:26.669833  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff32:ad4
03/26-10:17:42.584096  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff16:c89e
atharv@atharv:~/Documents/IS$
```

Part 4: Intrusion Analysis and Prevention

Step 7: Analyze IDS Logs

1. View Snort Alerts:

sudo tail -f /var/log/snort/alert

```
atharv@atharv:~/Documents/IS$ sudo tail -f /var/log/snort/alert
03/26-10:11:50.226882  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:11:50.678604  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff49:62bb
03/26-10:12:31.598159  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ffd7:f8e7
03/26-10:12:31.599387  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::16
03/26-10:12:31.816478  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::16
03/26-10:12:32.725350  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff5c:c4db
03/26-10:12:33.723483  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:13:26.297169  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {UDP} 0.0.0.0:68 -> 255.255.255.255:67
03/26-10:13:26.669833  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff32:ad4
03/26-10:17:42.584096  [**] [1:527:8] BAD-TRAFFIC same SRC/DST [**] [Classification: Potentially Bad Traffic] [Priority: 2] {IPV6-ICMP} :: -> ff02::1:ff16:c89e
^C
atharv@atharv:~/Documents/IS$
```

Step 8: Mitigate Vulnerabilities

1. Block Attacker's IP Address:

sudo ufw deny from 10.10.10.4

2. Harden SSH Configuration:

sudo nano /etc/ssh/sshd_config (Please change the following in the file

PermitRootLogin no AllowUsers test

3. Restart SSH:

sudo systemctl restart ssh

```
atharv@atharv:~/Documents/IS$ sudo ufw deny from 192.168.0.0
Skipping adding existing rule
atharv@atharv:~/Documents/IS$ sudo nano /etc/ssh/sshd_config
atharv@atharv:~/Documents/IS$ sudo systemctl restart ssh
atharv@atharv:~/Documents/IS$
```